Noesis: Selective Memory Injection for Cross-Tool AI Conversation Consistency

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https://github.com/AngryKiwiWithLegs/Noesis

August 2026

Abstract

Users converse with multiple AI assistants daily, yet each tool's memory is isolated: a stance expressed in one tool is unknown to the others, and each conversation starts from scratch. Existing approaches either store flat facts without a notion of confidence, or inject all retrieved context regardless of relevance, causing token costs to explode with memory size. We present **Noesis**, a local-first memory layer that (1) extracts typed *thoughts* from conversations across tools into a shared per-user store, (2) gates injection behind a confidence lifecycle in which thoughts mature from *tentative* to *settled* based on repetition, assertion strength, cross-tool corroboration, and contradiction signals, and (3) caps injected context with a token budget. Across three models (Gemini Flash, gemma3:4b, qwen2.5:3b) on 110 paired questions each, memory injection raises answer accuracy from 41.8% to 85.5%, 34.5% to 82.7%, and 37.3% to 62.7% respectively (all $p < 0.001$, McNemar's test). At 500 stored memories, Noesis injects 35× fewer tokens than a naive RAG baseline while maintaining rank-1 retrieval (MRR 0.90 vs. 0.002). Cross-tool retrieval reaches 50% where per-tool isolation is structurally bound to 0%. We release the system, experiment harness, and all raw results.

1 Introduction

Heavy users of AI assistants switch between ChatGPT, Claude, Gemini, and local models throughout the day. Each tool maintains its own isolated memory: a database preference carefully explained to Claude must be re-explained to GPT; a design decision reached with Gemini is invisible to Claude. The reasoning a user performs across tools—their positions, preferences, and open questions—evaporates at every tool boundary.

Two families of systems address parts of this problem, each with a critical gap. *Fact stores* such as Mem0 [2] persist extracted facts but treat all memories equally: a firmly held professional identity and a passing mention of a tool have the same standing, and there is no mechanism for a memory to earn trust over time. *Retrieval-augmented generation* (RAG) [5] retrieves relevant context for each query, but naive variants inject everything retrieved; as personal memory grows into the hundreds of items, injected context grows linearly, burying the relevant signal and inflating cost (we measure 7,192 tokens at n=500).

Noesis is built on a different premise: *what* gets injected matters as much as *whether* anything is injected. Thoughts should earn their way into the context the way beliefs earn their way into a mind—through repetition, consistency, and survival against contradiction.

This paper makes the following contributions:

1. **A confidence lifecycle for conversational memory** (§3). Thoughts are typed (identity, preference, position, question, synthesis, event, contradiction) and mature tentative→provisional→settled

via a four-signal scorer, with type-specific time decay and cluster-aware supersession that reduced stale-stance leakage from 93% to 7% (§3.3).

2. **Budget-capped selective injection that scales** (§5.3). At 500 memories Noesis injects ∼207 tokens (35× fewer than naive RAG) while holding MRR at 0.90; naive RAG's MRR collapses to 0.002 as targets are buried (§5.3).

3. **Quantified cross-tool memory sharing** (§5.4). With a shared per-user store, cross-tool retrieval reaches 50% (37% on strictly cross-tool queries) versus **0%** for per-tool isolation—a gap that is architecturally impossible for isolated systems to close.

4. **A replicated A/B result across model tiers** (§5.1). On identical n=110 question sets, memory injection significantly improves answer accuracy for a cloud model (Gemini Flash, +43.6pp), a mid-tier local model (gemma3:4b, +48.2pp), and a small local model (qwen2.5:3b, +25.5pp); all $p < 0.001$.

The system is local-first: embeddings and retrieval run on-device (sqlite-vec + all-MiniLM-L6-v2), and only optional LLM-based extraction calls a cloud API. All code, the experiment harness, raw results, and a verification script that recomputes every reported number from raw JSON are released.[1]

2 Related Work

Memory for LLM agents. MemGPT [7] treats the LLM as an operating system paging memory between contexts; Letta extends this with archival storage. Mem0 [2] extracts and updates atomic facts with an LLM pipeline, reporting gains over full-history baselines. These systems store *facts*; none models the *confidence* of a memory over time, and none addresses multiple front-end tools sharing one user's memory. Noesis differs in (a) its lifecycle, where injection rights are earned, and (b) its tool-agnostic proxy architecture, which makes any OpenAI-compatible client a first-class participant.

Retrieval-augmented generation. RAG [5] and successors retrieve per-query context. Self-RAG [1] and FLARE [4] improve retrieval quality or adaptivity, but address corpus QA, not a user's evolving personal memory. Our scale experiment (§5.3) shows why naive inject-everything retrieval fails for personal memory: cost grows linearly and ranking collapses. Noesis pairs hybrid retrieval (BM25 + vector + RRF, with time-decay weighting) with a hard token budget and a confidence gate.

Personalization. PALR [9] and related work personalize outputs from user interaction history, typically for recommendation-style tasks. Chat-tool vendor memory features (e.g., ChatGPT memory) are per-product and closed. Noesis's cross-tool experiment (§5.4) demonstrates a structural advantage of user-scoped rather than tool-scoped memory.

Visualization of embedding spaces. t-SNE [8] and UMAP [6] project high-dimensional embeddings for inspection; force-directed layouts [3] model attraction/repulsion in 2D. Our star map (§6) extends force-directed layout to 3D with a time-spring axis so that semantic clusters remain volumetric while recency reads as height.

[1] https://github.com/AngryKiwiWithLegs/Noesis

Table 1: Thought taxonomy with type-specific half-lives and injection priority.

Type	Definition	Half-life (days)	Priority
identity	Who the user is	∞	highest (always)
preference	Explicit preference	90	high
position	Stance on a topic	90	high
question	Open question under exploration	30	medium
synthesis	Insight from merged ideas	180	medium
event	Happened / decided	7	low (recent)
contradiction	Self-contradictory judgment	∞	flagged

3 System Design

Noesis sits between AI tools and model providers (Figure 1). Tools point at a local OpenAI-compatible proxy; the proxy injects memory into the system prompt, forwards the request to the configured provider (cloud or local), and captures the conversation for memory formation.

3.1 Thought taxonomy and two-phase capture

Conversational memory is typed into seven categories with distinct persistence semantics (Table 1). An *identity* thought ("I am a backend engineer") should outlive an *event* ("I tried Bun yesterday"), and the taxonomy encodes this as type-specific half-lives.

Capture is two-phase. *Phase 1* is synchronous and bounded (<15 ms): embed, deduplicate by hash, insert as *tentative*. The conversation never blocks on memory. *Phase 2* is asynchronous: an extractor (a small cloud LLM, or a deterministic fallback) classifies the thought; a confidence scorer updates its status; a supersession check retires contradicted predecessors; the cold store (Markdown + YAML frontmatter in an Obsidian vault) is updated for human readability and editability.

3.2 Confidence lifecycle

Every thought begins *tentative*: stored, but never injected and invisible to retrieval—the "capture threshold low, injection threshold high" principle. Four signals promote a thought: *repetition* (the same stance recurs across sessions), *assertion strength* ("I've decided" outranks "maybe"), *cross-tool consistency* (distinct phrasings of a stance from different tools), and *absence of contradiction*. *Provisional* thoughts inject when topic-relevant; *settled* thoughts always inject. Confidence evolves incrementally: reinforcement $+0.10\,(1 - C)$, conflict -0.20, and time decay $C \leftarrow C\,e^{-0.693\,\Delta t/t_{1/2}}$ with $t_{1/2}$ set by type (Table 1).

3.3 Cluster-aware supersession

When a user changes their mind ("I moved from MySQL to PostgreSQL"), the old stance should retire. Embedding similarity is the wrong signal here: contradictory stances mention different entities and therefore have *low* similarity. Noesis detects mind-changes by requiring jointly (1) a replacement verb ("switched to", "migrated to", …, EN+ZH), (2) the same domain via a keyword map (both stances are about *databases*), (3) stance-like type, and (4) injectable status of the new thought. Matched predecessors are marked *superseded*—excluded by every retrieval path—and linked via `superseded_by`. In our evaluation this reduced stale-stance leakage from 93% to 7% (Table 2).

Table 2: Supersession fix: stale-stance leakage before/after (n=30 scenarios).

Metric	Before	After	Δ
Old stance leaked into context	28/30 (93%)	2/30 (7%)	−86pp
New stance surfaced	12/30 (40%)	28/30 (93%)	+53pp
Ideal (new shown, old retired)	1/30 (3%)	26/30 (87%)	+84pp

3.4 Selective retrieval and injection

At query time, three channels run in parallel (Figure 2): BM25 for exact terms, dense retrieval for paraphrase, and two lightweight signals (recency, core-facts). Channels fuse via reciprocal-rank fusion with time-decay weighting. A confidence gate drops everything below *provisional*, and a hard token budget (1,200 tokens) truncates the remainder. The result is injected into the system prompt as compact structured lines.

4 Evaluation Setup

We evaluate four research questions:

- **RQ1**: Does memory injection improve answer accuracy? (A/B, end-to-end, §5.1)
- **RQ2**: Which components contribute? (ablation, retrieval-layer, §5.2)
- **RQ3**: How do cost and ranking scale with memory size? (§5.3)
- **RQ4**: Does cross-tool sharing work? (§5.4)

Models. Gemini Flash (cloud), gemma3:4b and qwen2.5:3b (local via Ollama), spanning capability tiers.

A/B design. 30 synthetic English user profiles, each with 5 build statements (identity/preference/position/event, followed by 5 test questions whose expected keyword is known from the build statements. For each question the model answers twice: *with memory* (via the Noesis proxy, which injects context) and *without memory* (direct to the provider). All models are evaluated on the identical first 110 questions (22 profiles). Scoring is objective keyword-hit; paired outcomes are tested with McNemar's χ^2.

Retrieval-layer experiments. Ablation, scale, and cross-tool experiments run against the real pipeline without LLM calls, measuring Recall@5, MRR, Precision@5, tentative leakage, and injected tokens.

Reproducibility. Every number in this paper is recomputed from raw JSON by a released verification script.

Table 3: A/B results on identical n=110 questions. All $p < 0.001$ (McNemar, df=1).

Model	n	**With**	**Without**	Δpp	χ^2
Gemini Flash	110	85.5%	41.8%	**+43.6**	42.48
gemma3:4b	110	82.7%	34.5%	**+48.2**	51.02
qwen2.5:3b	110	62.7%	37.3%	**+25.5**	16.57

Table 4: Ablation and baselines (50 queries, retrieval-layer).

Config	**Recall@5**	**MRR**	**Prec@5**	**Leak**
Noesis (full)	**100%**	**1.000**	**96%**	**10**
w/o core-fact signal	100%	1.000	96%	10
w/o recency signal	100%	1.000	96%	10
w/o confidence gating	100%	1.000	88%	**29**
w/o semantic retrieval	49%	0.160	100%	0
Naive RAG	100%	1.000	88%	29
Recent-window	22%	0.099	100%	0
Random	45%	0.206	94%	15

5 Results

5.1 RQ1: Memory injection improves accuracy across model tiers

Table 3 and Figure 3 show the headline result. Three findings stand out. First, *injection helps everywhere*: all three models improve significantly, and memory *hurt* in only 10 of 330 paired answers (3.0%). Second, *memory levels the field between local and cloud*: with injection, gemma3:4b (4B parameters, local) reaches 82.7%, nearly matching Gemini Flash's 85.5%, while its no-memory baseline (34.5%) trails Gemini's 41.8%. Third, *small models are comprehension-capped*: qwen2.5:3b receives the same injected context but reaches only 62.7%; inspection shows the memory was present in-prompt and the model failed to use it—a model-capability ceiling, not a retrieval failure.

5.2 RQ2: Semantic retrieval is the engine; gating is the moat

Removing semantic retrieval collapses recall (100%→49%), confirming it as the engine (Table 4). Removing the confidence gate leaves recall intact but leaks 29 tentative thoughts versus 10—gating is what keeps *unverified* material out of the prompt. Core-fact and recency signals show no measurable effect in this query-driven suite; their value emerges on identity-relevant but topically unrelated queries, which this suite undersamples (an honest gap).

5.3 RQ3: Cost stays flat while baselines explode

Figure 4 sweeps memory size from 10 to 500 with 10 known targets embedded among fillers. Noesis's injected cost is flat (144→207 tokens) because the budget caps it by construction, and MRR holds at 0.90—the target sits at rank 1. Naive RAG keeps recall only by injecting everything: 7,192 tokens at n=500, with the target buried near rank 500 (MRR 0.002). Recent-window recall collapses to 0% beyond n=10: the window forgets everything older. The methodological takeaway: *recall alone is misleading*; rank and token cost must be reported together.

5.4 RQ4: Cross-tool sharing vs. structural isolation

In 40 scenarios a stance stated under tool A is queried under tool B (Figure 5). Noesis retrieves 50% overall and 37% on strictly cross-tool queries; the isolated baseline retrieves **0%** on the latter—not because it ranks poorly, but because the memory is not in its partition. This is the structural argument: per-tool memory systems cannot share by design, and no amount of per-tool tuning closes the gap. Noesis's misses trace to retrieval precision (the memory exists but ranks below top-k), not to sharing.

6 The Star Map Visualization

The stored memory graph is visualized as a galaxy (Figure 6). Positions are not assigned by axes; they emerge from a force simulation: *semantic attraction* pulls embedding-similar thoughts together in 3D, *repulsion* prevents overlap (with an inter-cluster regime that keeps galaxies separated), a weak *time spring* biases height by recency, and *semantic Z-adhesion* keeps related thoughts from being sheared apart in height. Topic constellations are therefore *discovered* by the physics rather than imposed—the visual counterpart of the clustering the retrieval pipeline exploits. The viewer is a local web tool shipped with the system.

7 Discussion

When memory helps most. Gains concentrate on questions that are unanswerable without personal context (job, stack, stances). On mainstream questions both groups often answer correctly, understating the effect—our hit rates are lower bounds on the true gap.

Weaker models benefit disproportionately. gemma3:4b posts the largest relative gain (+165% over its baseline). Memory injection is therefore most valuable exactly where model capability is scarce—small local models—which matters for privacy-sensitive and offline settings.

Retrieval precision is the current bottleneck. The cross-tool misses, several A/B misses, and the supersession fix's residual 7% all reduce to ranking failures, not architecture failures. Improving ranking (rerankers, query expansion) is the highest-leverage next step.

Honest negatives. We report that supersession was originally dead code leaking stale stances 93% of the time, that core-fact and recency signals showed no effect in our ablation suite, and that a 3B model is comprehension-capped despite correct injection. We consider these findings part of the contribution.

8 Future Work

Thought branching. When a new judgment deviates from history by more than a threshold ($\delta = 1 - \cos > 0.45$), Noesis should create a *branch* linked via `evolved_from` rather than superseding, preserving the full trajectory of a stance. This enables metrics we cannot compute today: stabilization depth, branching rate, and post-branch divergence—a longitudinal record of how a user's positions evolve.

Structured injection. Replacing flat context lines with a structured `user_thought_graph` (anchor topic, trajectory with weights, cross-links) may improve the ability of small models to exploit injected memory, directly addressing the qwen2.5 comprehension ceiling.

Stronger evaluation. LLM-as-judge scoring to complement keyword hits; distractor questions; a second cloud model to replicate the Gemini result; longitudinal deployment data beyond synthetic profiles.

9 Limitations

Scoring is keyword-hit, which is objective but coarse; semantic correctness may differ in both directions. Personas are synthetic; build/test alignment may overstate real-world gains, and real conversations are noisier. The cloud A/B uses one provider (Gemini Flash); n=110 rather than the planned 150 due to free-tier rate limits (reported transparently; all significance thresholds are exceeded by wide margins). Retrieval-layer experiments measure what reaches the prompt, not final answer quality. The supersession detector relies on replacement verbs and a domain keyword map; paraphrased mind-changes outside its lexicon will be missed.

10 Conclusion

Noesis shows that *what* a memory system injects matters as much as *whether* it injects. A confidence lifecycle earns injection rights through repetition and consistency; a token budget keeps cost flat as memory grows (35× cheaper than naive RAG at n=500 with better ranking); and a shared per-user store makes cross-tool consistency possible where per-tool architectures are structurally bound to fail. The result is a local-first layer that makes a 4B local model answer personal questions as well as a frontier cloud model—and that keeps the user's reasoning in the user's own vault.

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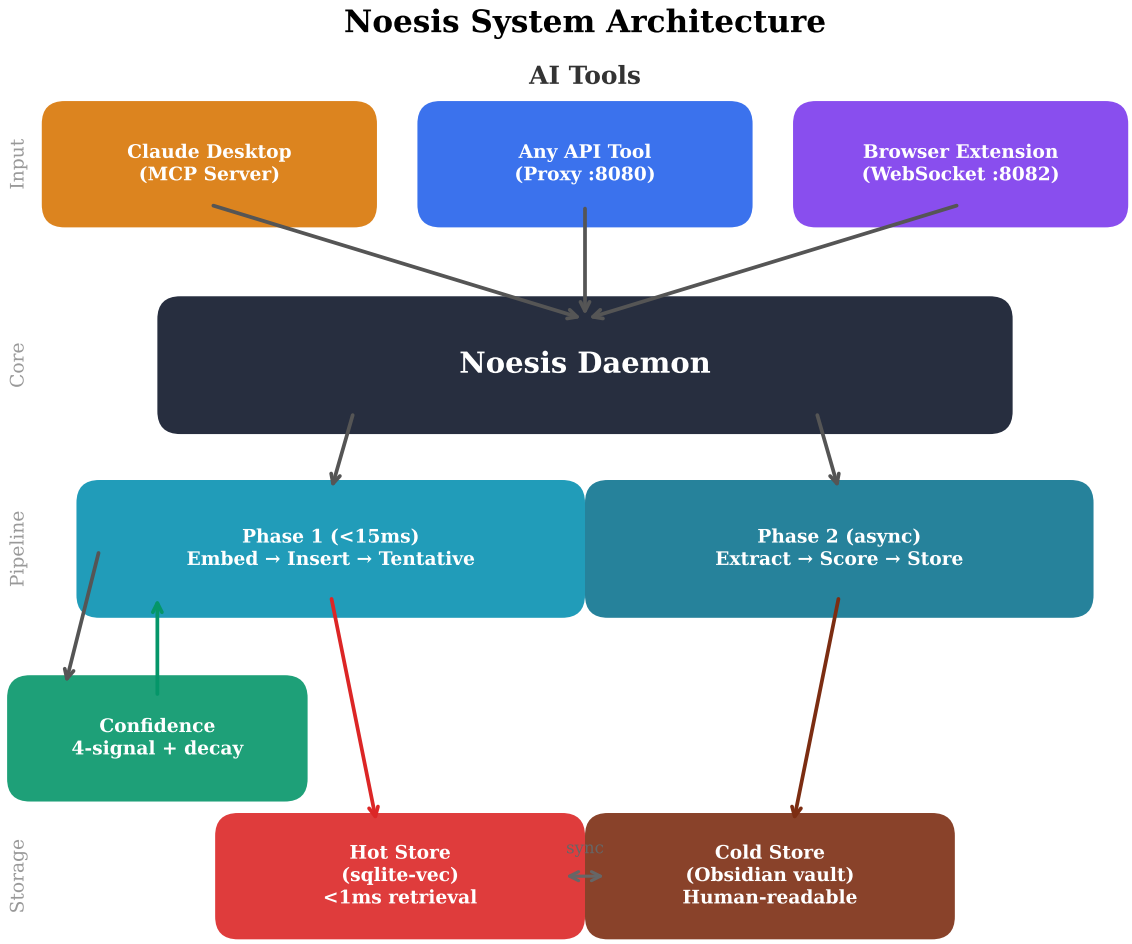


Figure 1: Noesis system architecture. Any OpenAI-compatible tool routes through the local proxy; Phase 1 stores the utterance synchronously (<15 ms) and Phase 2 asynchronously extracts, scores, and links the thought. Hot store (sqlite-vec) serves retrieval; cold store (Obsidian vault) keeps memory human-readable and editable.

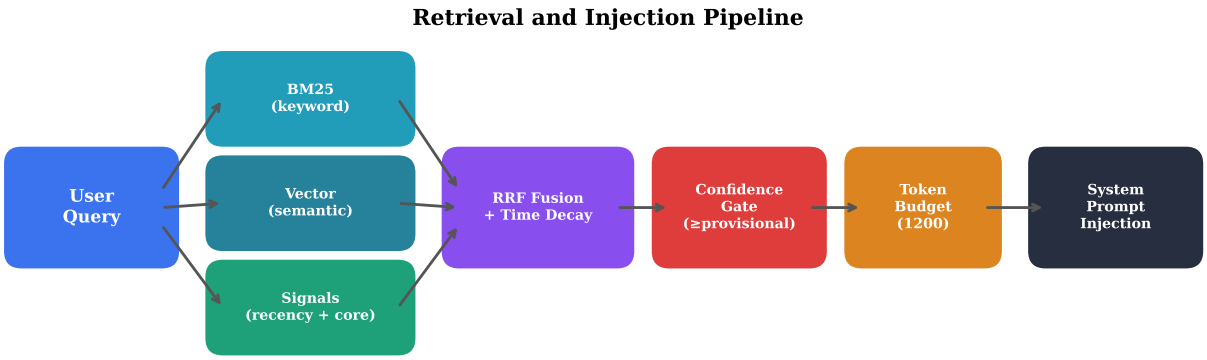


Figure 2: Retrieval and injection pipeline. Three channels fuse via RRF; a confidence gate removes non-injectable thoughts; a token budget caps injected context.

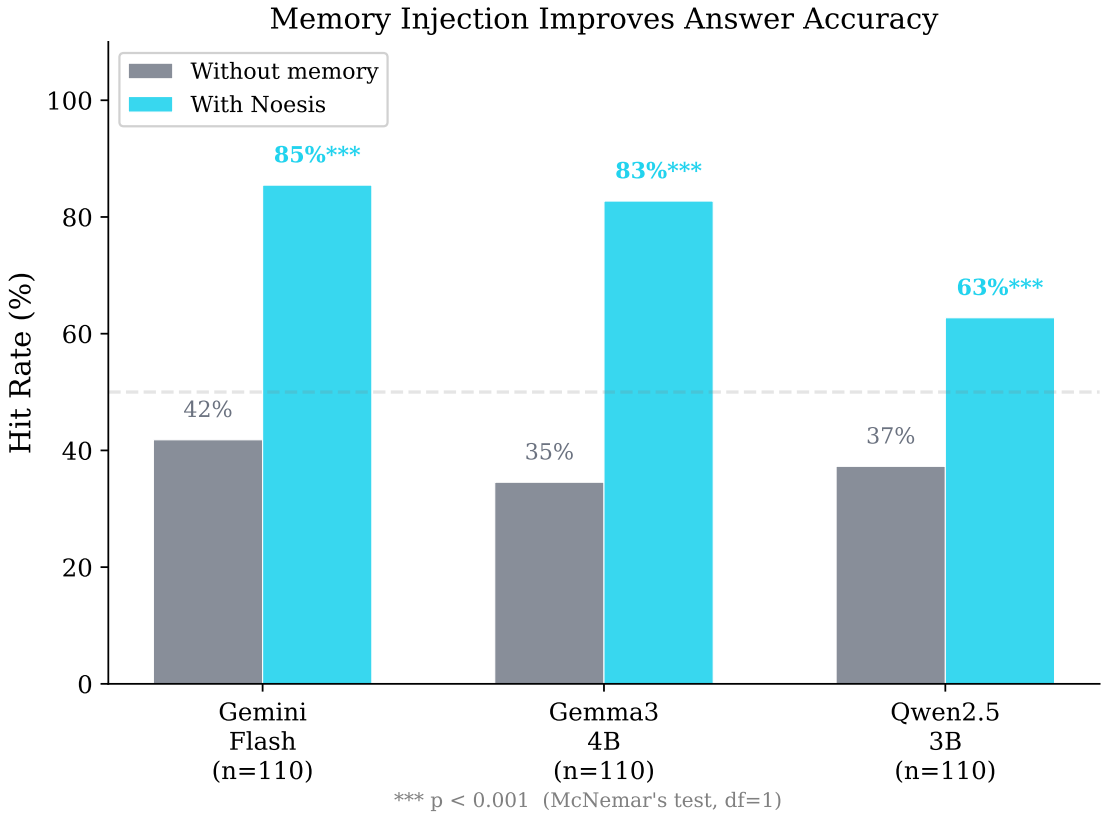


Figure 3: A/B accuracy across models (n=110 each). All improvements significant at $p < 0.001$.

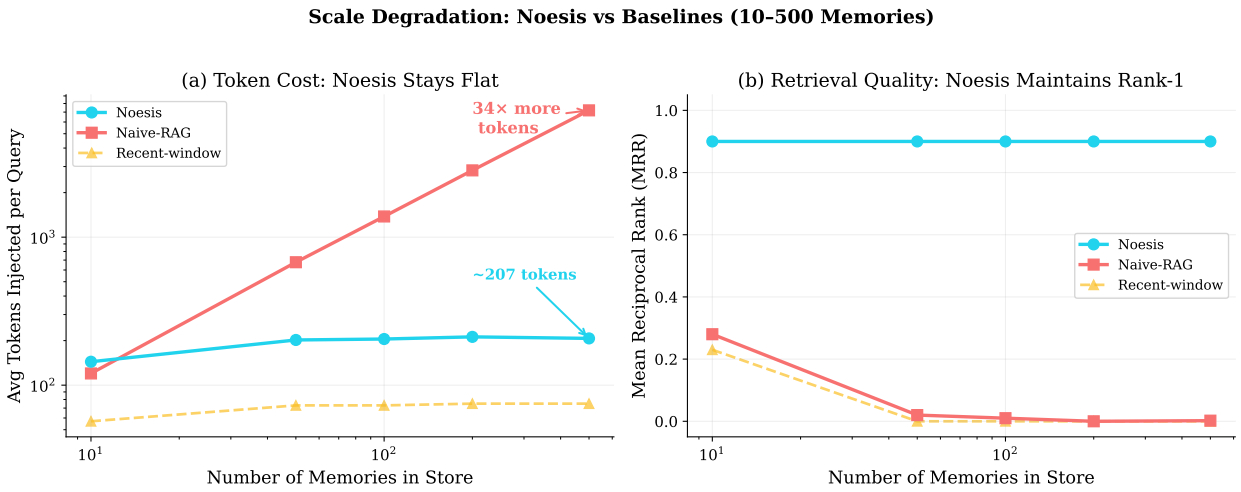


Figure 4: Scale behavior from 10 to 500 stored memories. (a) Noesis's injected tokens stay flat (∼207) while naive RAG grows to 7,192 (35×). (b) Noesis holds MRR at 0.90; naive RAG collapses to 0.002 as targets are buried; recent-window loses recall entirely past n=10.

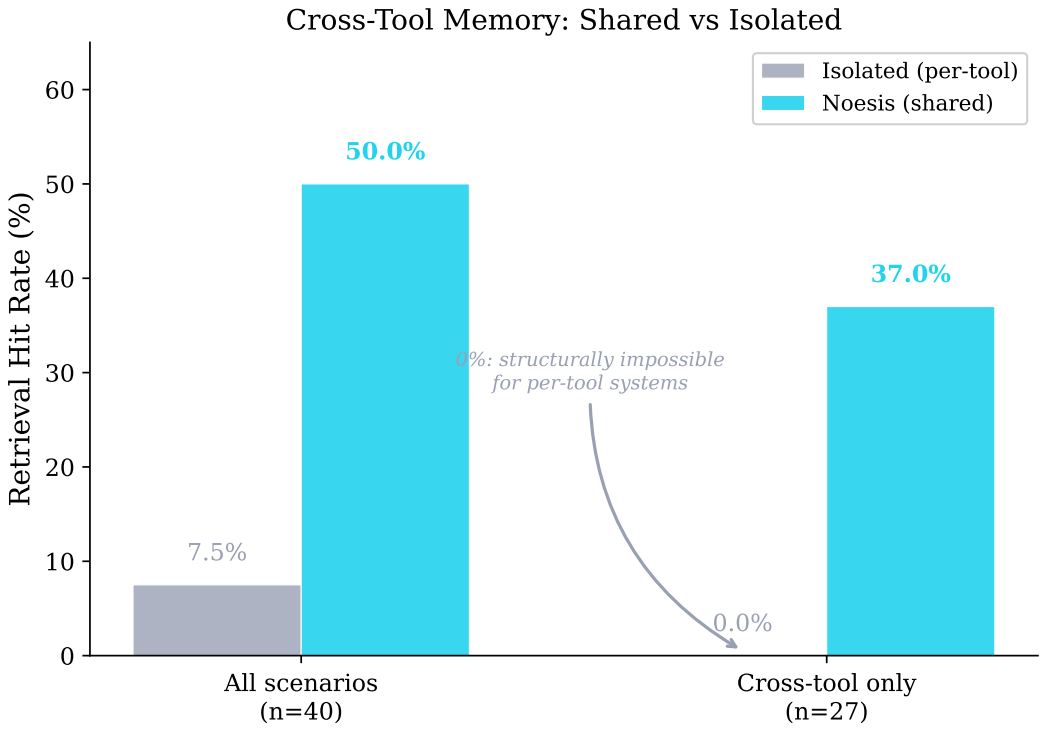


Figure 5: Cross-tool retrieval. Isolated per-tool memory scores 0% on strictly cross-tool queries by construction.

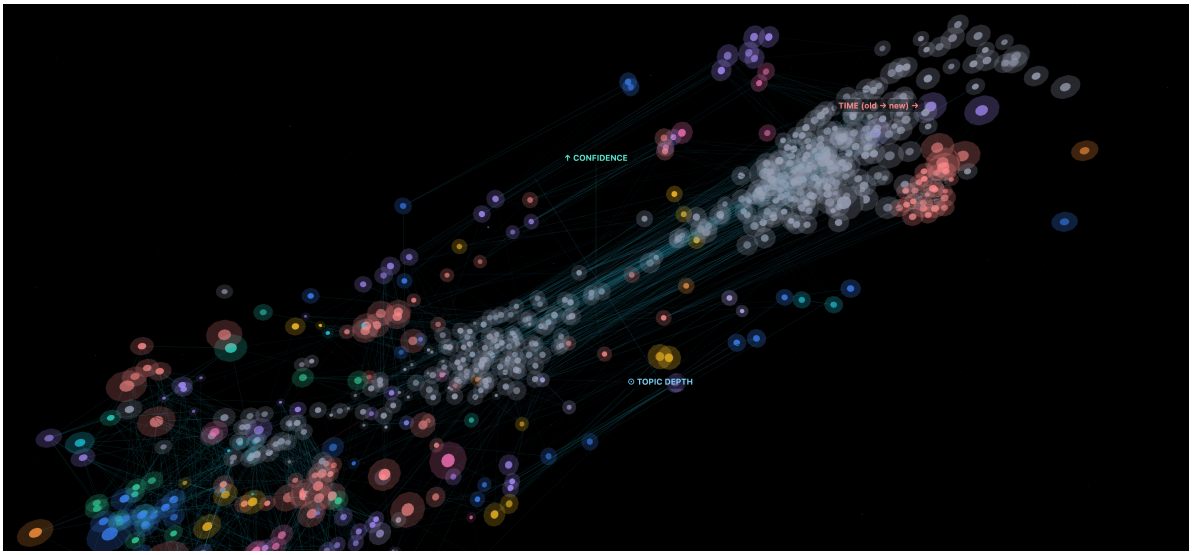


Figure 6: The star map. Semantic neighbors attract into volumetric constellations; a weak time spring biases height so newer thoughts (higher) separate from older ones without tearing clusters apart. Color encodes topic; size encodes type and activation weight.